Making Wage Rules Legible: A RAG Chatbot for Minimum-Wage Workers in Metro Manila

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***Abstract*—This paper identifies a focused research direction for a future Retrieval-Augmented Generation (RAG) chatbot addressing statutory wage-benefit awareness among minimum-wage workers in the National Capital Region (NCR). Following the narrowing framework of Real Problem → Relevant SDG → Narrow Problem → Research Question → Data Sources → Possible Language Model, this paper selects and justifies SDG 8 (Decent Work and Economic Growth), narrows a broad labor-rights problem to a single, bounded information need, and formulates a primary research question that examines the chatbot's ability to provide accurate, evidence-supported answers while minimizing hallucinations. The paper also identifies three publicly available knowledge-base sources, presents an initial evaluation data plan, and proposes a preliminary language model with justification. It concludes with a reflection on why the selected problem is sufficiently narrow and manageable for a student research project.**

Keywords — *retrieval-augmented generation, SDG 8, statutory monetary benefits, hallucination, labor rights*

I. Selected UN SDG and Justification

Selected SDG: SDG 8 — Decent Work and Economic Growth. Relevant SDG area: Target 8.8, which calls to "protect labor rights and promote safe and secure working environments for all workers, including migrant workers, in particular women migrants, and those in precarious employment." Justification: Nearly one in five establishments in the Philippines fail basic labor standards inspections, and almost half fall short on occupational safety requirements [1]. Two-thirds of Filipino workers remain in informal employment with little or no protection, and the share of workers in more protected formal positions actually fell in 2025 relative to 2024, moving further from the government's own target [2]. This is not a settled problem but a worsening one. Minimum-wage workers are among those least equipped to verify whether they are receiving their correct statutory entitlements, because the governing rules are not contained in a single accessible source. Connection to RAG: A chatbot that retrieves and synthesizes information from authoritative wage-rule documents could allow a worker to ask a plain-language question and receive an answer grounded in the actual governing text, rather than requiring the worker to locate and interpret the relevant provision unaided. This directly operationalizes Target 8.8's concern with protecting workers in precarious employment.

II. Narrow Problem Statement

Level	Group Output
Broad domain	Labor and employment rights in the Philippines
SDG-related issue	Lack of accessible information on statutory labor protections, undermining decent work (SDG 8, Target 8.8)
Target users	Rank-and-file, minimum-wage, land-based, non-agricultural private-sector workers employed in the National Capital Region (NCR)
Narrow problem	These workers cannot easily determine whether they are receiving correct minimum wage, overtime pay, holiday pay, and 13th month pay, because the governing rules are scattered across the Labor Code, a DOLE statutory-benefits handbook, and a regional wage order, each using different terminology and updated on different schedules
Proposed chatbot focus	E.g., "What is the current minimum wage in NCR for a small retail establishment?"; "Am I entitled to overtime pay for work beyond 8 hours?"; "How is 13th month pay computed?"

Table I. Problem-Narrowing Path

Problem statement: Rank-and-file, minimum-wage workers employed by private-sector, non-agricultural establishments in the National Capital Region frequently cannot determine whether they are receiving their correct statutory monetary benefits. This difficulty arises because the rules governing minimum wage, overtime pay, holiday pay, and 13th month pay are distributed across three distinct sources — the Labor Code, a DOLE-issued benefits handbook, and a regional wage order, each written for a different audience and updated on a different schedule. This problem is worth investigating because it is measurable (a worker's entitlement can be checked against a known set of rules), bounded (restricted to one region and one category of benefit), and addressable through a defined, obtainable document collection, rather than requiring case-by-case legal interpretation. This project explicitly excludes overseas Filipino workers (governed by the Department of Migrant Workers and POEA rules), kasambahay/domestic workers (governed by the separate Batas Kasambahay), and termination or dismissal disputes (a distinct and more fact-dependent legal area), in order to keep the knowledge base and evaluation set within reach of a student research project.

III. Primary Research Question

Research Question: How effectively can a RAG-based chatbot provide accurate, evidence-supported answers while minimizing hallucinations in NCR statutory monetary-benefit questions?

Justification: This research question focuses on how well a RAG-based chatbot can answer statutory monetary-benefit questions within the defined NCR scope. In particular, the study examines whether the chatbot can provide accurate answers that are supported by the authoritative labor documents retrieved from its knowledge base while avoiding factual or legal claims that are not supported by those documents. The question is measurable because the generated responses can be evaluated against prepared reference answers and their corresponding supporting passages. For this study, a hallucination refers to a factual or legal claim in a generated response that cannot be supported by the retrieved passages from the defined knowledge base. This keeps the study focused on the quality and reliability of the chatbot's responses rather than simply determining whether the system can be developed.

IV. Initial Knowledge-Base Data Sources

	Source	Doc Type	Relevance	Availability	Reliability
1	Labor Code of the Philippines — DOLE, 2022 Ed.	Statutory text	Statutory basis for wage, overtime, holiday, and 13th month pay	dole.gov.ph, free	Authoritative primary legal source
2	Handbook on Workers' Statutory Monetary Benefits — DOLE-BWC, 2024 Ed.	Reference handbook	Plain-language computation guidance for worker-facing answers	bwc.dole.gov.ph, free	Issued by DOLE's wage-standards unit
3	Wage Order No. NCR-27 — NWPC, eff. July 2026	Regional wage order	Current binding minimum wage rates for NCR	nwpc.dole.gov.ph, free	Official NWPC regulatory issuance

Table II identifies three initial knowledge-base sources, exceeding the minimum of two required.

These three sources, jointly cover the statutory, practical, and current-rate dimensions of the wage rules relevant to NCR workers: the Labor Code establishes the legal basis, the BWC handbook translates it into computation guidance, and Wage Order No. NCR-27 fixes the current binding rate. Together they form a fixed, small, and fully public document collection suitable for a bounded student knowledge base.

V. Initial Evaluation Data Plan

No evaluation data has yet been collected. The following describes how a preliminary test question set would be constructed in a future phase of this study.

Source of questions: Realistic questions would be drafted by the research group based on common wage and benefit concerns documented in DOLE's own public advisories and inspection findings, rather than sourced from actual workers, since no data collection has yet occurred.

Question types: The set would include easy questions with directly stated answers (e.g., the current NCR minimum wage rate), moderate questions requiring combining information across two documents (e.g.,

computing 13th month pay from a stated monthly rate), and difficult questions requiring the correct provision to be distinguished from a similarly worded but inapplicable one (e.g., the difference between regular holiday and special non-working day pay rules). The evaluation set will contain approximately 30–50 realistic questions within the defined scope of NCR statutory monetary benefits. The questions will vary in difficulty, including straightforward questions that can be answered directly from a single provision, moderate questions that require identifying the appropriate rule, and more difficult questions that may require information from multiple relevant passages. The questions will focus on areas such as minimum wage, overtime pay, holiday pay, and 13th month pay.

Establishing reference answers: Expected answers and their supporting passages will be manually identified and verified against the three source documents in Table II before the chatbot evaluation takes place. This provides a consistent basis for checking the generated responses without relying on the chatbot's own output to determine whether an answer is correct or supported.

Evaluation approach: The generated responses will be evaluated against the prepared reference answers and their corresponding supporting passages from the knowledge base. The evaluation will examine whether the chatbot provides the expected answer and whether the factual or legal claims in its response are supported by the retrieved information. Claims that cannot be supported by the retrieved passages will be identified as hallucinations for the purpose of this study. This allows the evaluation to consider both the accuracy of the generated answers and the extent to which the information provided by the chatbot can be traced back to the authoritative documents used by the system.

Evaluation metrics: Two measures will be considered in evaluating the chatbot's responses: answer accuracy and unsupported claim rate. Answer accuracy will assess whether the generated response agrees with the prepared reference answer. Unsupported claim rate will be used as the study's measure of hallucination by determining the proportion of factual or legal claims in a generated response that cannot be supported by the retrieved passages. Together, these measures will evaluate both the correctness of the chatbot's answers and the extent to which its generated claims are supported by the authoritative information retrieved from the knowledge base.

Measurement procedure: Answer accuracy will be assessed using a three-level scoring rubric. A score of 2 will be assigned when the generated response is correct and its essential information agrees with the prepared reference answer; a score of 1 will be assigned when the response is partially correct but contains missing or inaccurate essential information; and a score of 0 will be assigned when the response is incorrect or fails to provide the expected answer. The overall answer accuracy score will be calculated by dividing the total points obtained by the maximum possible points and multiplying the result by 100.

Unsupported claim rate (UCR) will be calculated by identifying the factual or legal claims in each generated response and checking whether each claim can be supported by the retrieved passages. The number of unsupported claims will be divided by the total number of factual or legal claims and multiplied by 100. A lower UCR indicates fewer unsupported claims or hallucinations in the generated responses.

VI. Possible Language Model and Justification

Model name: A Groq-hosted, open-weight, instruction-tuned large language model (e.g., openai/gpt-oss-20b or a comparable model), consistent with the model class used in prior coursework for this class.

Justification: An open-weight instruction-tuned model of this size provides a practical option for evaluating mixed formal English and Taglish questions that Filipino workers may realistically use. Its accessibility through a hosted API also makes repeated evaluation across a bounded question set more manageable for a resource-limited student study. The model is therefore proposed as an initial option whose performance can be examined during the future evaluation rather than assumed in advance.

Expected role: The model would receive a worker's natural language question together with the top-K retrieved passages from the knowledge base in Table II, and generate an answer restricted to that retrieved context, including a defined fallback response when no relevant provision is retrieved.

Practical considerations: The model is accessed via a hosted API, requiring no local GPU. Free-tier usage is expected to be sufficient for a bounded 30–50 question evaluation set. This model selection is preliminary: it is proposed on the basis of accessibility and suitability for the task, not on comparative benchmarking against alternative models, and may be revisited once a full study is undertaken.

VII. Group Reflection

This problem is sufficiently narrow and manageable for a student research project for four reasons. First, it is restricted to a single region: NCR contains no provinces, so wage rates vary only by sector and establishment size rather than by geography, avoiding the added complexity a multi-region study would introduce. Second, it is restricted to a single category of entitlement, statutory monetary benefits, explicitly excluding termination and dismissal

disputes, which depend on facts a document-based chatbot cannot verify. Third, the knowledge base is fixed and small: three publicly available documents with no access or permission barriers, rather than an open-ended or continuously expanding document collection. Fourth, the research design centers on one primary and clearly measurable evaluation of the RAG chatbot's responses. The study focuses on whether the chatbot can provide accurate, evidence-supported answers while minimizing hallucinations, rather than introducing multiple model or retrieval comparisons. This keeps the evaluation focused and manageable within the timeframe of a student research project.

VIII. References

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